Thursday 22/03/2022

MODULE-3

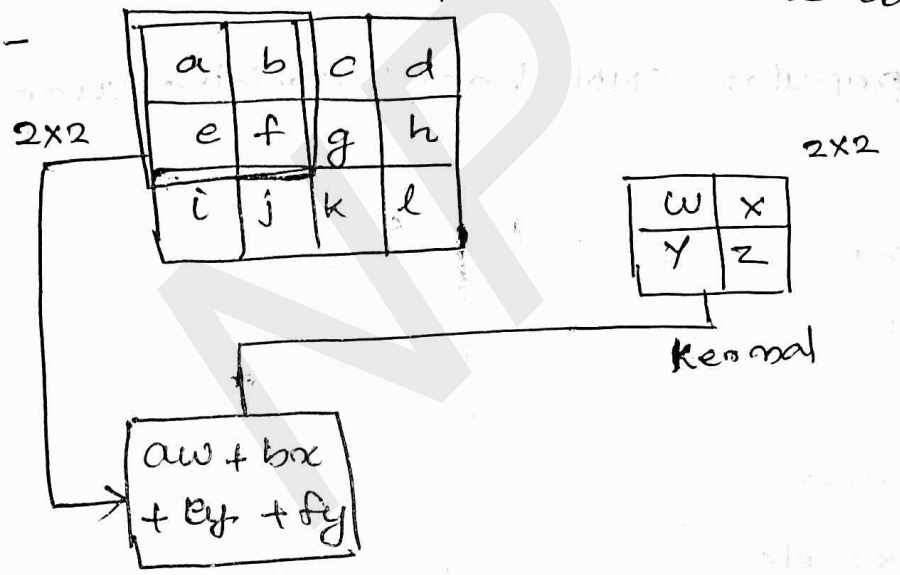
CNN

* CNN is a specialized Neural network used for processing Grid like topology data

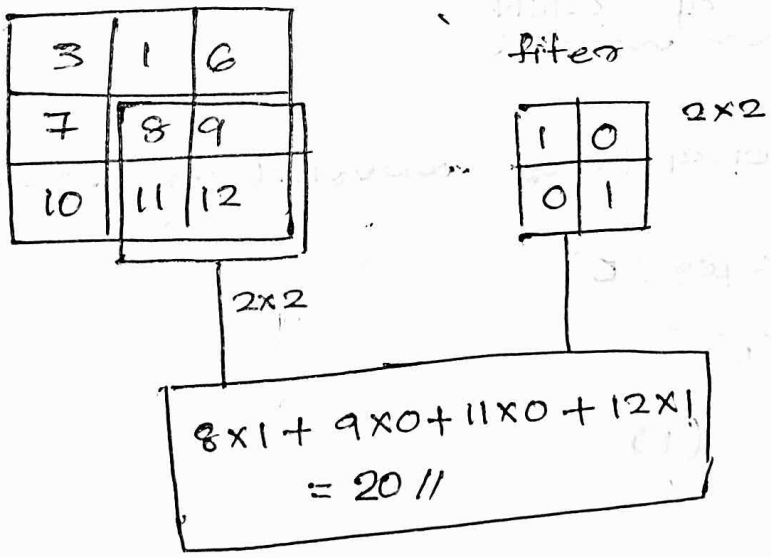
example → Image.

* CNN uses convolution operation on the data

example –



Eg:–



CNN is defined as the networks that use convolution operation in atleast one layer.

Applications of CNN

* It is used to extract features in the data
* It can be used for classification task.
* It can be used for segmentation task.

CNN Based Architectures

Different popular CNN based architectures are

- Lenet
- Alexnet
- ZFnet
- VGG
- Googlenet.
- Resnet etc.

Basic structure of CNN

A CNN consists of convolution layer (C)

- Convolution layer (C)
- ReLu Layer (R)
- Pooling Layer (P)

(i) Convolution Layer

* Convolution layer performs convolution operations on the input data

(ii) ReLu Layer

ReLu layer apply activation function

(iii) Pooling Layer

Pooling Layer performs pooling operation.

pooling 3 types
- Max pool
- Average pool
- Min pool

* Convolution and ReLu layer usually stack together after that pooling layer appears

Basic patterns of CNN are shown below.

- Convolution ReLu Convolution ReLu Pooling

CR CR P

-

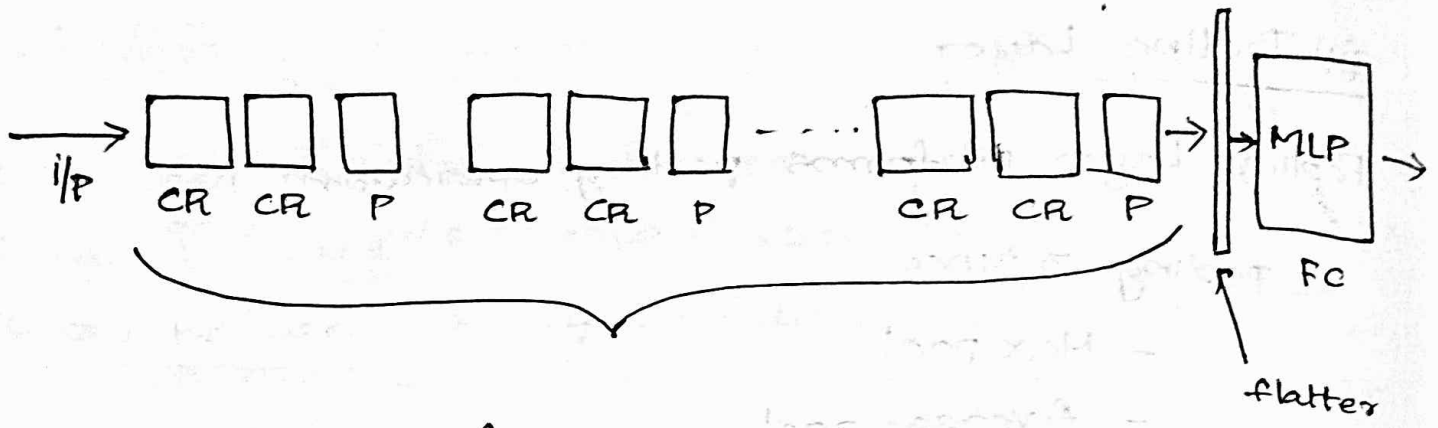
CR P CR P . . . .

* To create a deep neural network that performs a classification task the above pattern is repeated and at the end a fully connected layer FC is attached.

example

CR CR P CR CR P . . CR P FC

An FC layer is specifically designed to ~~solve~~ perform classification task



CONVolution

Convolution is a linear dot product operations A convolution operation places a filter (kernal) at each possible position of the image and performs a convolution operation on that positions. The result is a feature map

data

1	2	3	4
5	6	7	8
9	10	0	1
2	1	2	3

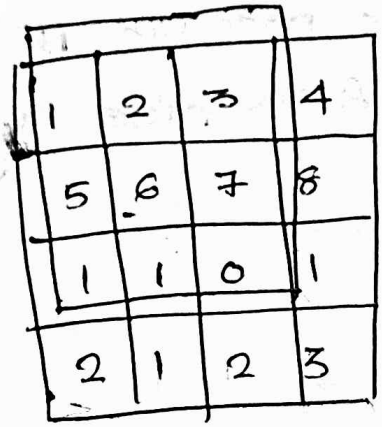
4×4

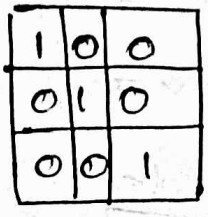
filter

1	0	0
0	1	0
0	0	1

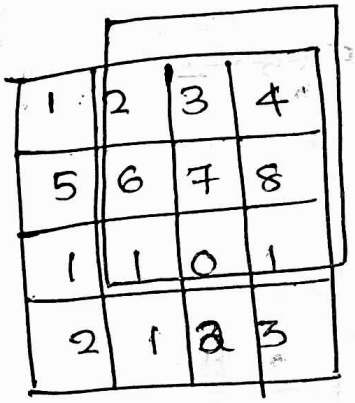
Findout the feature map after the convolution Operation

[Assume stride = 1]

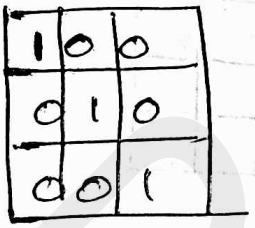




$1\times1+6\times1 = 7$



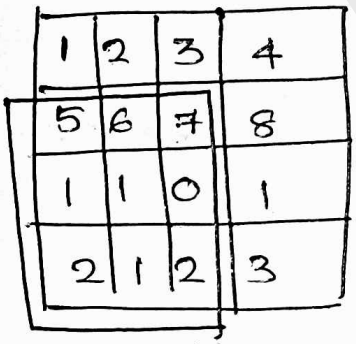
filter

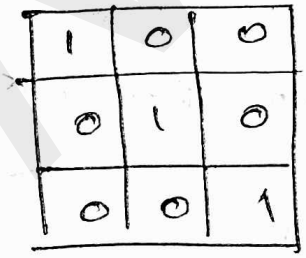


$2\times1 + 7\times1 + 1\times1$

$= 2+7+1$

$= 10$

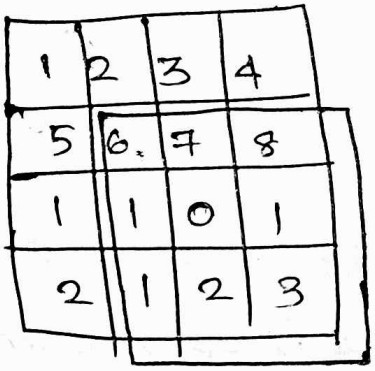


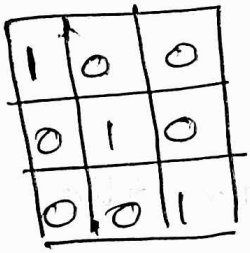


$5\times1 + 1\times1 + 2\times1$

$= 5+1+2$

$= 8$



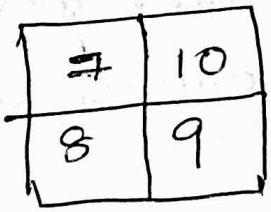


$6\times1 + 0\times1 + 3\times1$

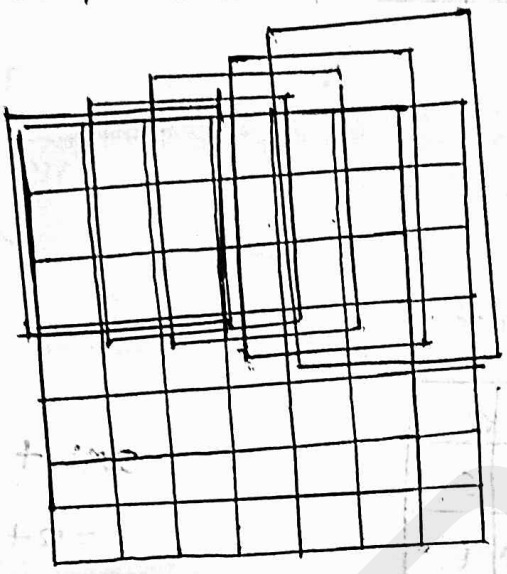
$= 6+0+3$

$= 9$

→



Q) How many spatial alligments possible, if the input size is 7x7x1 and filter size is 3x3x1. With stride = 1



~~5x5x2~~

5x5x1

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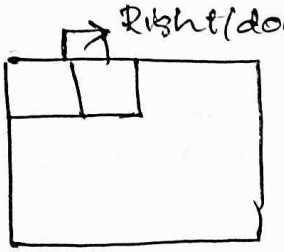
Stride

Stride is a parameter that determines the position filter should move.

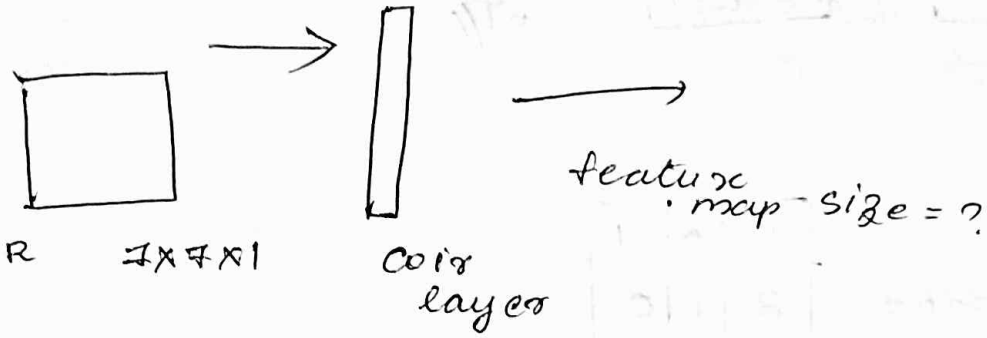


eg:-

* When stride = 1, the filter moves one position to the right after performing convolution operation.
* When stride = 2, the filter moves two position to the right after performing convolution operation.

Q) Findout the feature map size for below

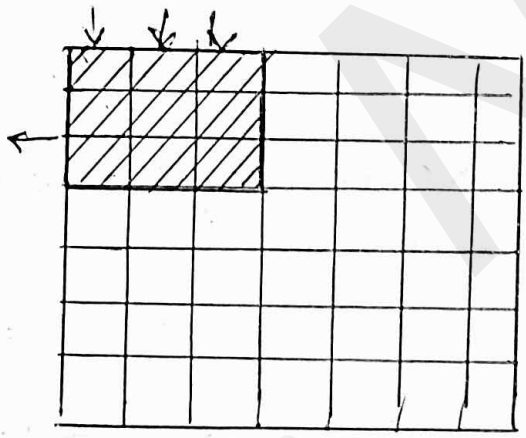
i/p size : 7 x 7 x 1



filter size = 3x3x1

Stride = 2.

padding = none.



Q) Findout feature map generated

1	1	1	1	1	1	0
0	0	0	1	0	0	0
1	0	0	1	0	0	0
0	1	0	1	0	0	0
1	0	1	0	0	1	0
1	1	0	0	0	0	0
1	1	1	1	1	1	0

filter

1	0	0
0	1	0
0	0	1

3x3

Stride=2
padding
=none

1	1	1
0	0	0
1	0	0

*

1	0	0
0	1	0
0	0	1

$= 1\times1 + 1\times0 + 1\times0 + 0\times0 + 0\times1$
$0\times0 + 1\times0 + 0\times0 + 0\times1$
$= 1$//

Feature map =

1	2	1
3	1	0
3	2	0

1	1	1
0	0	0
0	0	0

*

1	0	0
0	1	0
0	0	0

$= 1\times1 + 1\times0 + 1\times0 + 0\times0 +$
$1\times1 + 0\times0 + 0\times0$
$= 2$//

1	1	0
0	0	0
0	0	0

*

1	0	0
0	1	0
0	0	0

$= 1\times1 + 1\times0 + 0\times0 + 0\times0 + 0\times1$
$+ 0\times0 + 0\times0$
$= 1$//

1	0	0
0	1	0
1	0	1

*

1	0	0
0	1	0
0	0	0

$= 1\times1 + 0\times0 + 0\times0 + 0\times0 + 1\times1 +$
$0\times0 + 1\times0 + 0$
$= 3$//

0	1	0
0	1	0
1	0	0

*

1	0	0
0	1	0
0	0	0

$= 0\times1 + 1\times0 + 0\times0 + 0\times0 + 1\times1 +$
$0\times0 + 1\times0 + 0\times0 + 0\times0$
$= 1$//

0	0	0
0	0	0
0	1	0

*

1	0	0
0	1	0
0	0	0

$= 0\times1 + 0\times0 + 0\times0 + 0\times0 + 0\times0 +$
$0\times1 + 0\times0 + 0\times0$
$= 0$//

size of the feature map

Let i/p size is nxn, filter size = fxf, stride = s : then,

Size of the feature map =

$$= \left(\frac{n-f}{s}+1\right) * \left(\frac{n-f}{s}+1\right)$$

Padding

Padding describes the addition of empty pixels all around the ~~models~~ borders of the image. The purpose of padding preserve the original size of the image while doing convolution operations on the image and perform full convolution on the edge pixels.

eg:-

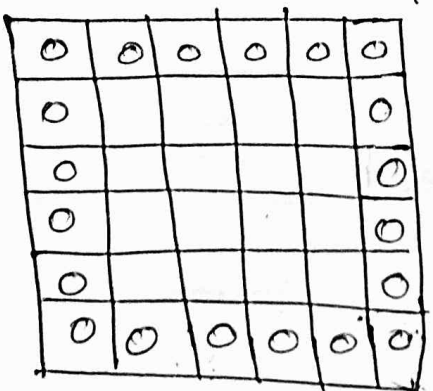
Consider an image of size 4×4

filter size = 2×2.

Stride = 1

padding = 1

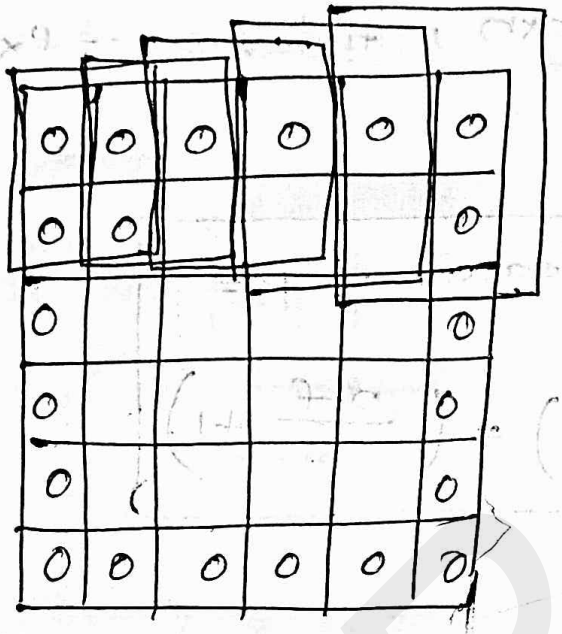
findout the feature map size.



padded image

if convolution performed with padding = 1

then feature size = 5×5



In general, while padding $\frac{(f-1)}{2}$ pixels is added all around the borders of an image..

As a result the spatial height and width is increased by (f-1).

* Suppose image size = n×n, feature size = f×f stride = s, padding = p, Then, the ~~feature m~~ size of feature map generated.

$$= \left(\frac{n+2p-f}{s}+1\right) * \left(\frac{n+2p-f}{s}+1\right)$$

Types of padding

(1) Valid padding

When padding is not used

(ii) Half padding

$\frac{(f-1)}{2}$ pixels with 0s added all around borders.

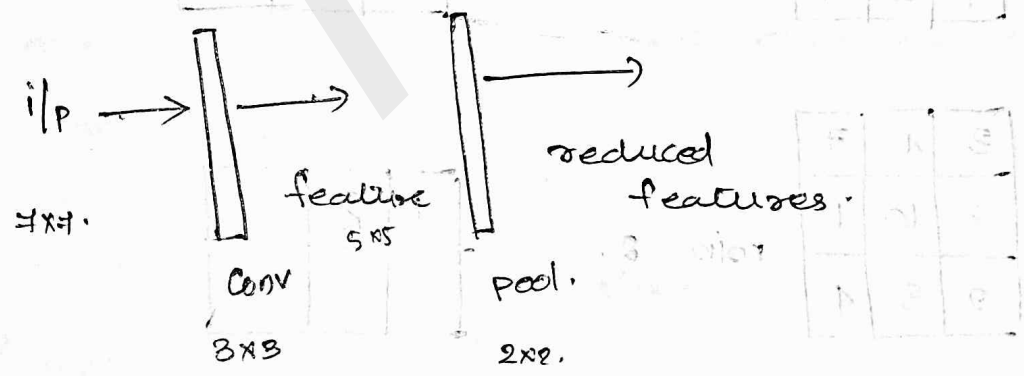
(iii) Full padding

(f-1) pixels w –

padding is 'same' if the output feature map is same as the input size

Pooling layer & Pool operation

Pooling layer is used to reduce the feature map size



Types of pooling

→ max poding

→ min pooling.

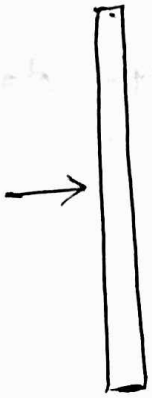
→ Average pooling.

Example :–

Pooling on the feature map.

6	3	4	7	8
9	8	10	11	12
1	2	3	4	5
6	7	8	9	10

feature map



pool

size = 3x3.

Type = Min Pool

pooling size = 3x3

Type = Min pool.

Stride = 1

6	3	4
9	8	10
1	2	3

min = 1

1		

3	4	7
8	10	11
2	3	4

min = 2.

1	2	

4	7	8
10	11	12
3	4	5

min = 3

1	2	3

9	8	10
1	2	3
6	7	8

min = 1

1	2	3
1		

8	10	11
2	3	4
7	8	9

min = 2

1	2	3
1	2	0

10	11	12
3	4	5
8	9	10

min = 3

1	2	3
1	2	3

* Pooling operation works on small grid regions of size. ~~and fi~~ PXP and find maximum or minimum or average values in that grid.

The above process is repeated over the image with stride s.

* Size of the feature map after

If s=1.

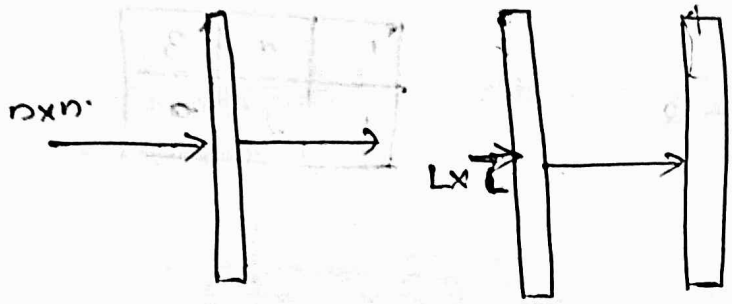
$$(L-P+1) * (L-P+1)$$

where,

L → I/P feature size

If s > 1

$$\left(\frac{L-P}{s}+1\right) * \left(\frac{L-P}{s}+1\right)$$

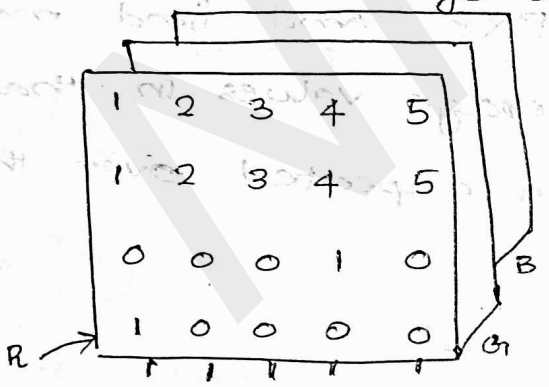


The o/p of convolution layer is feed into a ReLu layer.

Each feature value in the feature map

Tuesday 28/03/2023 Tutorial

Q) Given an RGB image of size 5×5×3



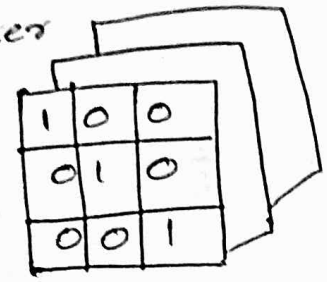
G

1	0	1	1	0
0	0	1	1	1
1	1	1	1	1
0	0	1	0	1
1	1	0	1	1

B

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

filter



1	0	0
0	1	0
0	0	1

3×3

Find the feature map generated after RGB is applied to a CNN (Stride = 1)

R channel

1	2	3	4	5
1	2	3	4	5
0	0	0	1	0
1	0	0	0	0
1	1	1	1	1

1	0	0
0	1	0
0	0	1

• 1×1 + 2×1 + 1×0
= 1 + 2
= 3//

2×1 + 3×1 + 1×1 = 2+3+1 = 6//

3×1 + 4×1 + 0×1 = 3+4+0 = 7//

1×1 + 0×1 + 0×0 = 1//

2×1 + 0×0 + 0×0 = 2//

3×1 + 1×1 + 0×1 = 4//

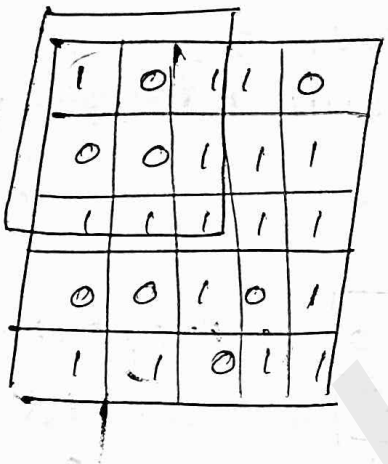
0×1 + 0×1 + 1×1 = 1//

0×1 + 0×1 + 1×1 = 1

0×1 + 0×1 + 1×1 = 1//

3	6	7
1	2	4
1	1	1

a channel



1	0	0
0	1	0
0	0	1

$1\times1 + 0\times1 + 1\times1 = 2$

$0\times1 + 1\times1 + 1\times1 = 0+1+1 = 2$

$1\times1 + 1\times1 + 1\times1 = 1+1+1 = 3$

$0\times1 + 1\times1 + 1\times1 = 0+1+1 = 2$

$0\times1 + 1\times1 + 0\times1 = 1$

$1\times1 + 1\times1 + 1\times1 = 1+1+1 = 3$

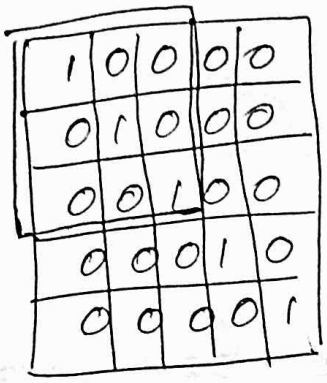
$1\times1 + 0\times1 + 0\times1 = 1$

$1\times1 + 1\times1 + 1\times1 = 3$

$1\times1 + 0\times1 + 1\times1 = 1+1 = 2$

2	2	3
2	1	3
1	3	2

B channel



1	0	0
0	1	0
0	0	1

$1\times1 + 1\times1 + 1\times1 = 3$

$0\times1 + 0\times1 + 0\times1 = 0$

$0\times1 + 0\times1 + 0\times1 = 0$

$0\times1 + 0\times1 + 0\times1 = 0$

$1\times1 + 1\times1 + 1\times1 = 3$

$0\times1 + 0\times1 + 0\times1 = 0$

$0\times1 + 0\times1 + 0\times1 = 0$

$0\times1 + 0\times1 + 0\times1 = 0$

$1\times1 + 1\times1 + 1\times1 = 3$

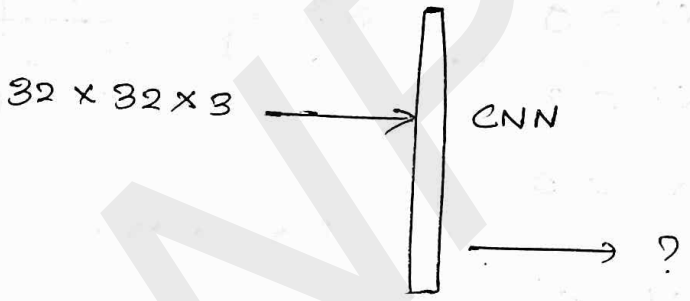
3	0	0
0	3	0
0	0	3

Feature map

3+2+3	6+2+0	7+3+0
1+2+0	2+1+3	4+3+0
1+1+0	1+3+0	1+2+3

8	8	10
3	6	7
2	4	6

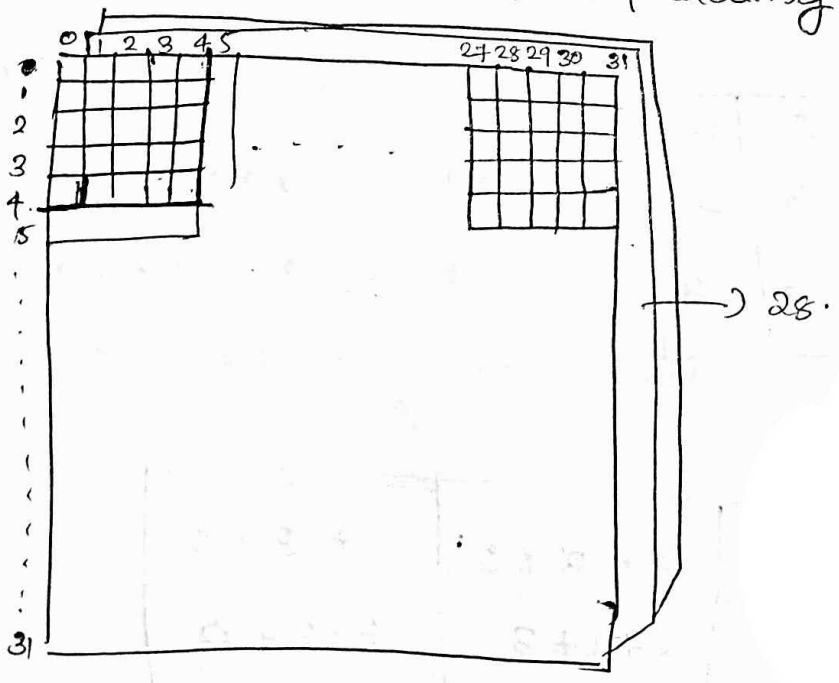
2) You have an input volume of size $32 \times 32 \times 3$ what are the dimensions of resulting volume after convolving a $5 \times 5 \times 3$ kernel with zero padding, stride = 1



stride = 1

5 x 5 x 3

zero padding



32 x 32

general equation

$$\frac{n-f}{s}+1 \times \frac{n-f}{s}+1$$

$s \rightarrow$ stride

$n \rightarrow$

$f \rightarrow$

$$\frac{n-f+1}{s} \times \frac{n-f+1}{s}$$

when $s=1$,

$$\frac{n-f}{s}+1 \times \frac{n-f}{s}+1$$

$$= \frac{32-5}{1}+1 \times \frac{32-5}{1}$$

$$= (32-5+1) \times (32-5+1)$$

∴ final feature map size

~~28 × 28~~ $28 \times 28 \times 1$

---

when $s=2$.

~~for~~ $\frac{n-f}{s}+1 \times \frac{n-f}{s}+1 = \frac{32-5}{2}+1 \times \frac{32-5}{2}+1$

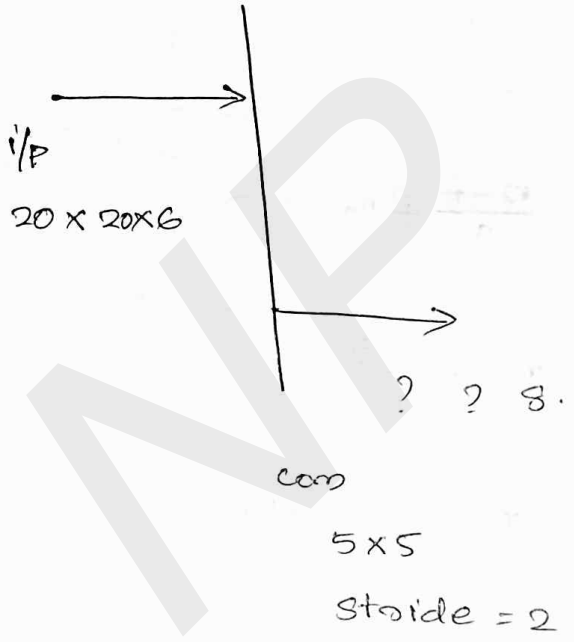
$$= \frac{32-5+1}{2} \times \frac{32-5+1}{2}$$

$$= 14 \times 14$$

∴ final feature map size

$14 \times 14 \times 1$ //

HW q) consider a convolution layer. The input consists of six feature maps of size 20×20. The output consists of 8 feature maps and filters are of size 5×5. The convolutions is done with stride of 2 and zero padding. Findout the output feature map size



Thursday
30/03/2023

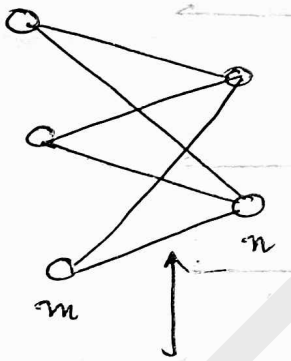
CNN → Sparse Interactions, Parameter sharing, Equivariant representation

* A CNN shows below properties
  1. Sparse interactions
  2. Parameter sharing
  3. Equivariant Representation.

1. Sparse interactions

* In a traditional neural network every output unit interact with every input unit

eg:-



no. of parameter = m x n

* A CNN has sparse connectivity
* By using the sparsee connectivity concept we can limit the number of connections.

For example :-

Q) consider a 4X4X1

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

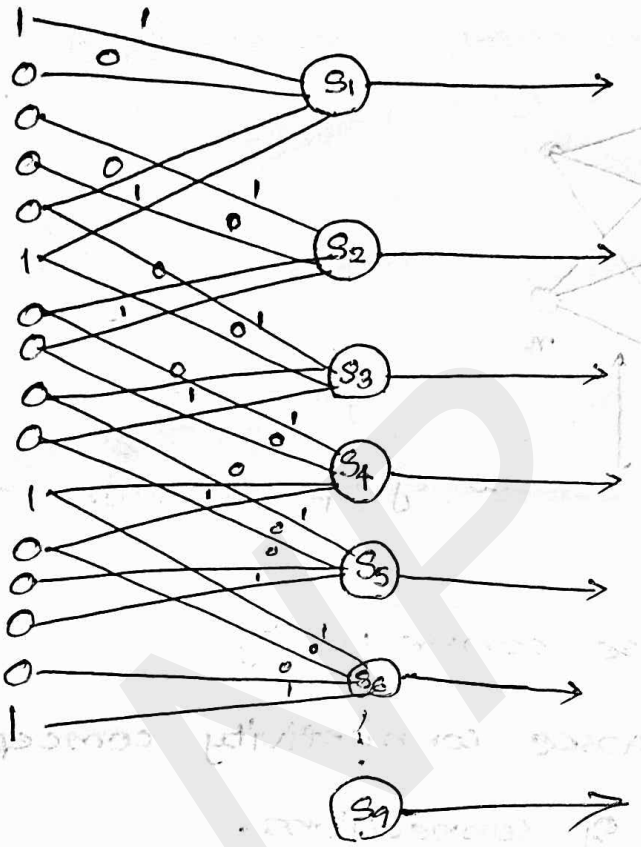
4X4X1

and a filter 2X2

1	0
0	1

construct a CNN ~~layer~~ Model using the above information with Stride = 1

flattened the matrix



Sparse connection

Here,

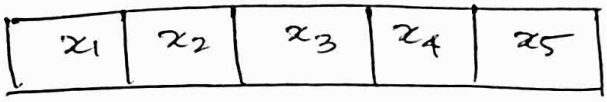
the number of connections

$= 9 \times 4$

$= 36$

* If we use a traditional Neural network. Then the number of connections needed will be greater than the CNN

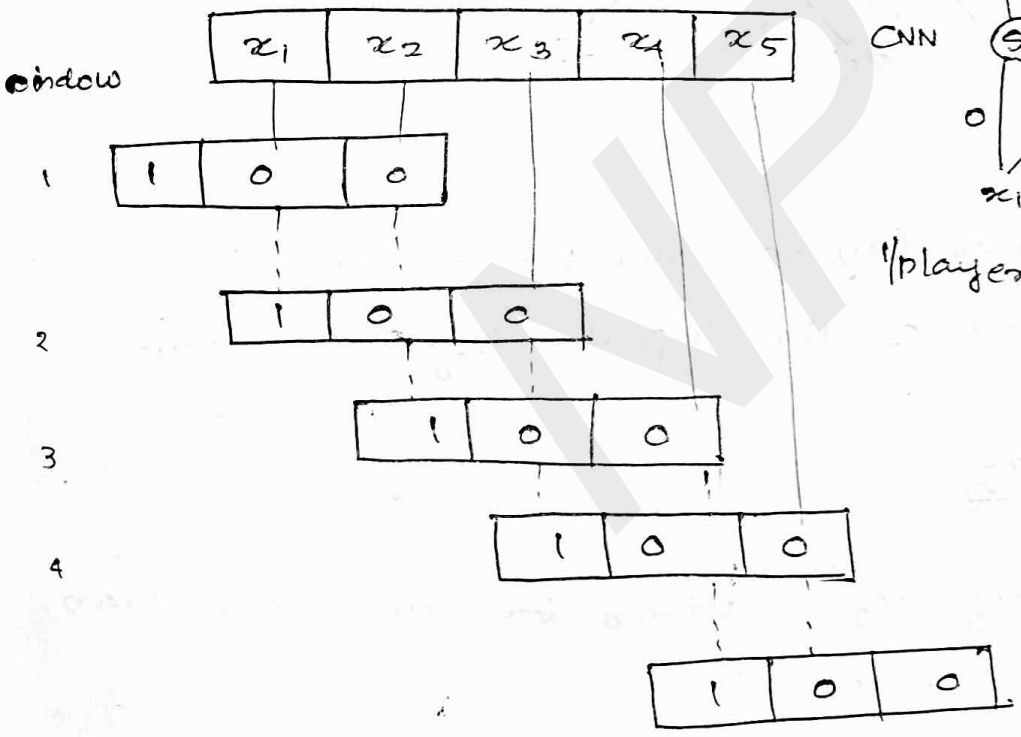
Consider an input array of one dimension

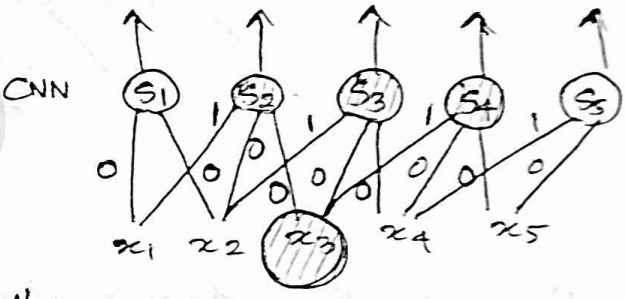


and considers a window of width = 3

Construct a CNN model of the above configuration with stride = 1

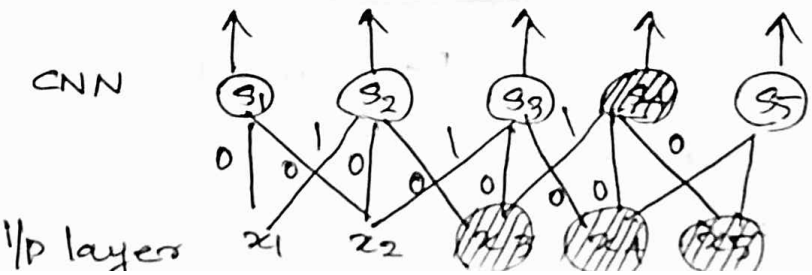
Ans





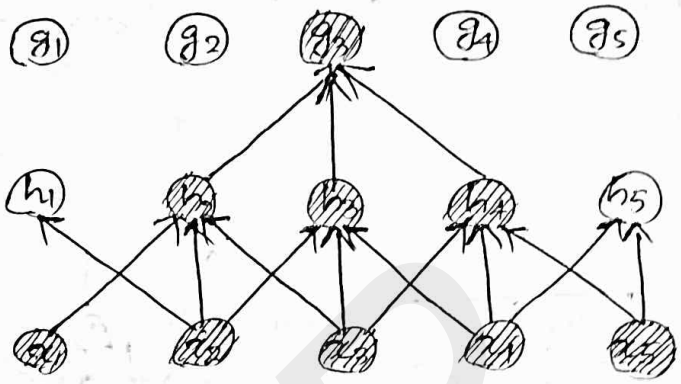
In this case number of connections is number of connections is 13

Receptive field : Receptive field of S_4 is



Receptive fields of the units in the deeper layer of a CNN is larger than receptive field of units in the shallow layers.

eg:-



* Units in the deeper layers can be indirectly connected to most of the inputs ~~layers~~ neurons

Parameter Sharing

- Parameter sharing refers using ~~the~~ for more than one function in the model.

eg:-

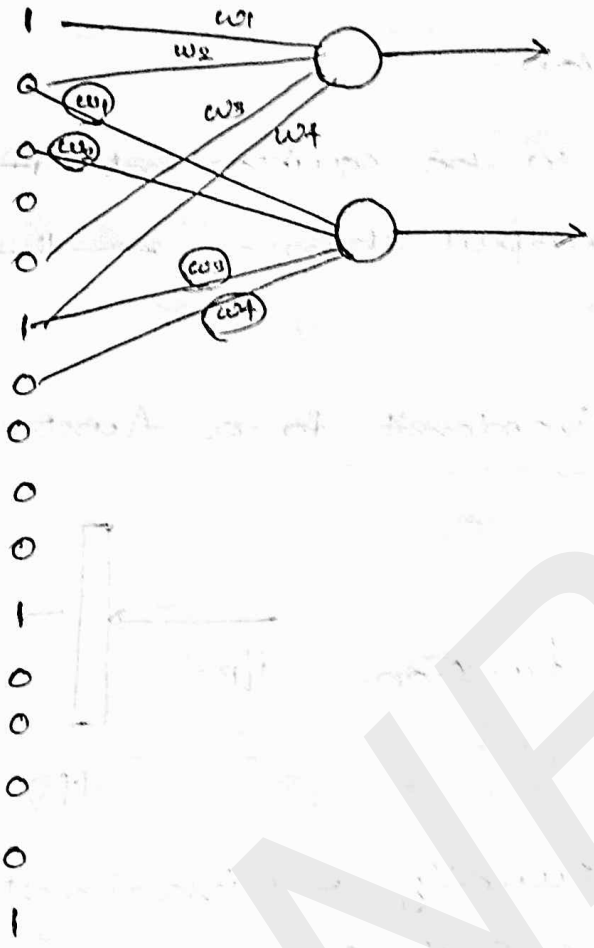
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

w_1	w_2
w_3	w_4

Edge detector

Corresponding CNN model.

(parameter sharing)



* A CNN automatically find parameters values that learns to perform edge detection etc. in the input image.

* No seperate set of parameters required at different locations to detect edges

CNN ഉപയോഗിക്കുമ്പോൾ parameters കുറയുന്നതിന്റെ കാരണം

Neural Network ⇒ $9 \times 4 = 36$

ADVANTAGES

* A CNN reduces the memory requirements and Statistical efficiency.

Equivariance

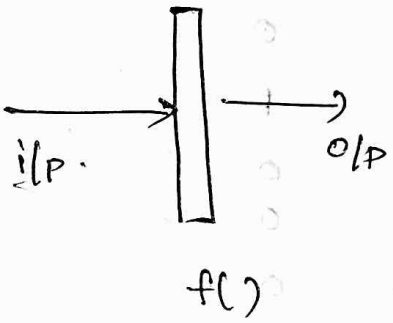
Equivariant Representation.

A function is said to be equivariant if the input changes then the output changes. ~~its the~~

A CNN is equivariant

A function $f(x)$ is equivariant to a function g if $\boxed{f(g(x)) = g(f(x))}$

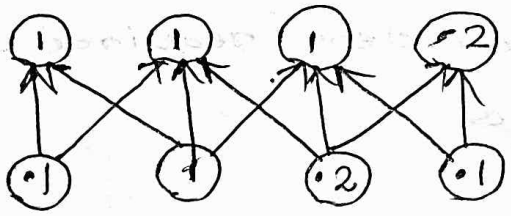
$g \rightarrow$ transformation function (shift function)



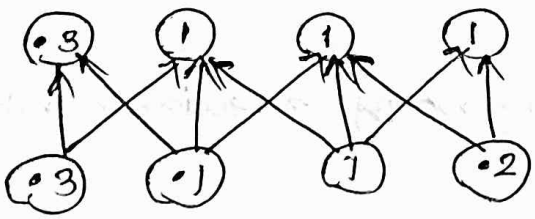
* Convolution is not naturally equivariant to other transformations such as
  → change in Scale rotation.

Pooling and Invariance

Pool:



Pool:

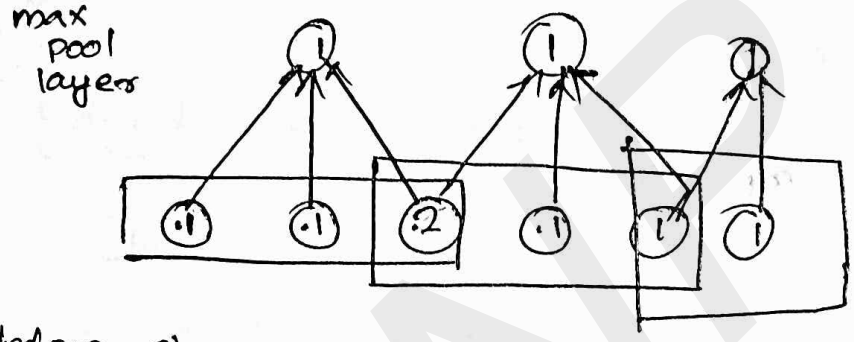


in the bottom figure input has been shifted to right by one pixel.

Every value in the input is changed after the shift but only half of the values in the pooling layer is changed.

08/04/23 Monday

Pooling and down Sampling



window size = 3
stride = 2

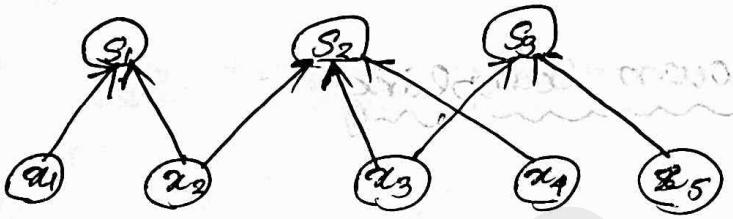
Here, we use max pooling with a pool width of is 3 and the stride between the pools equal to 2. This reduces the representation size by the factor of 2

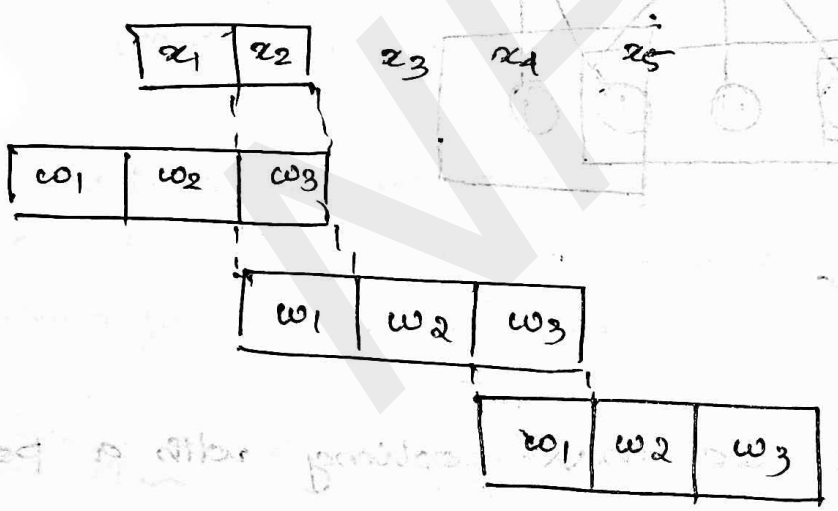
The pooling operation reduces the computational cost by reducing the number of parameters to the next layer.

Variants of basic convolution

1. Strided convolution

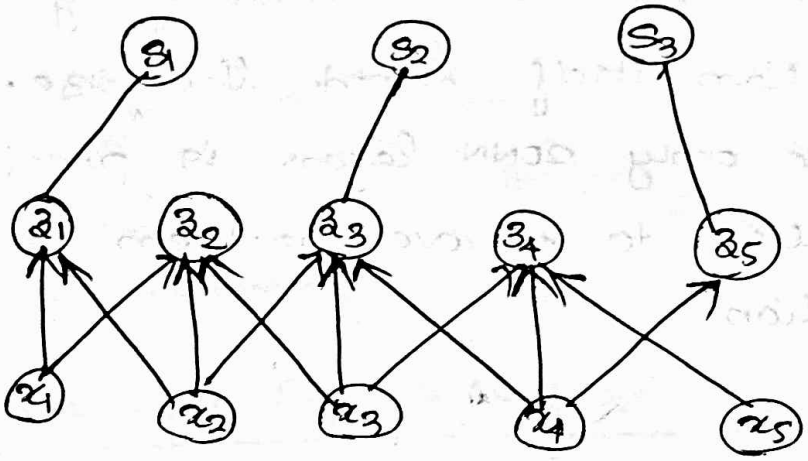
It is possible to use strides of different sizes to perform convolution for example stride = 2/3 etc.





* Convolution with a stride greater than one pixel is mathematically equivalent to convolution with unit stride followed by down sampling.

* But the two step approach involving unit stride followed by down sampling is computationally wasteful.



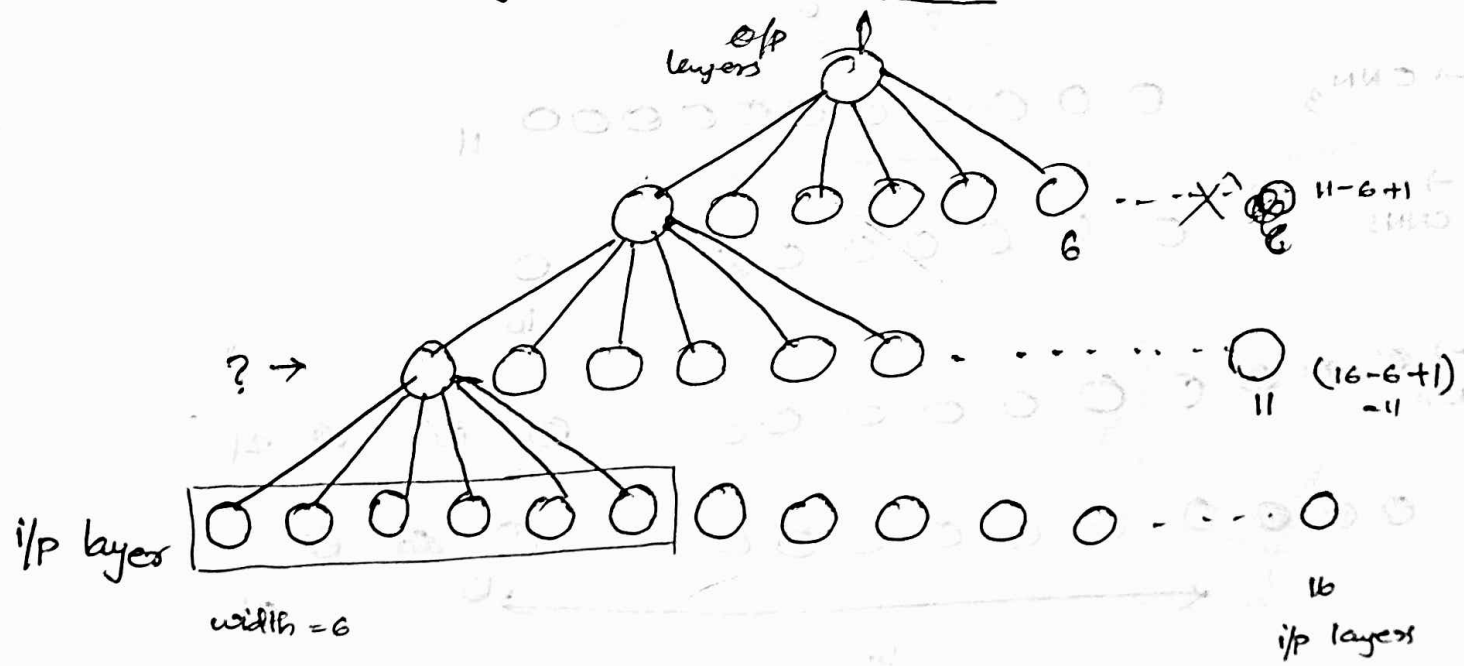
Tuesday
04/04/2023

Padded convolution

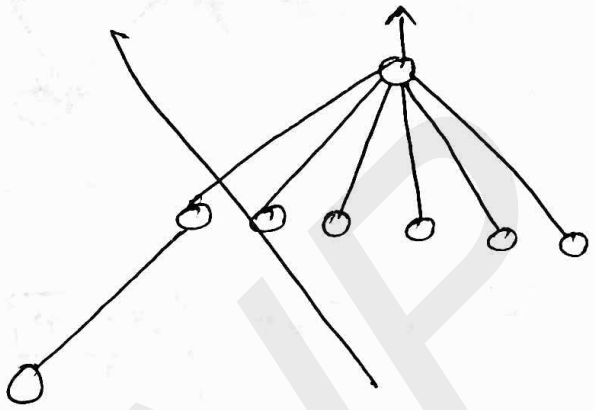
Types

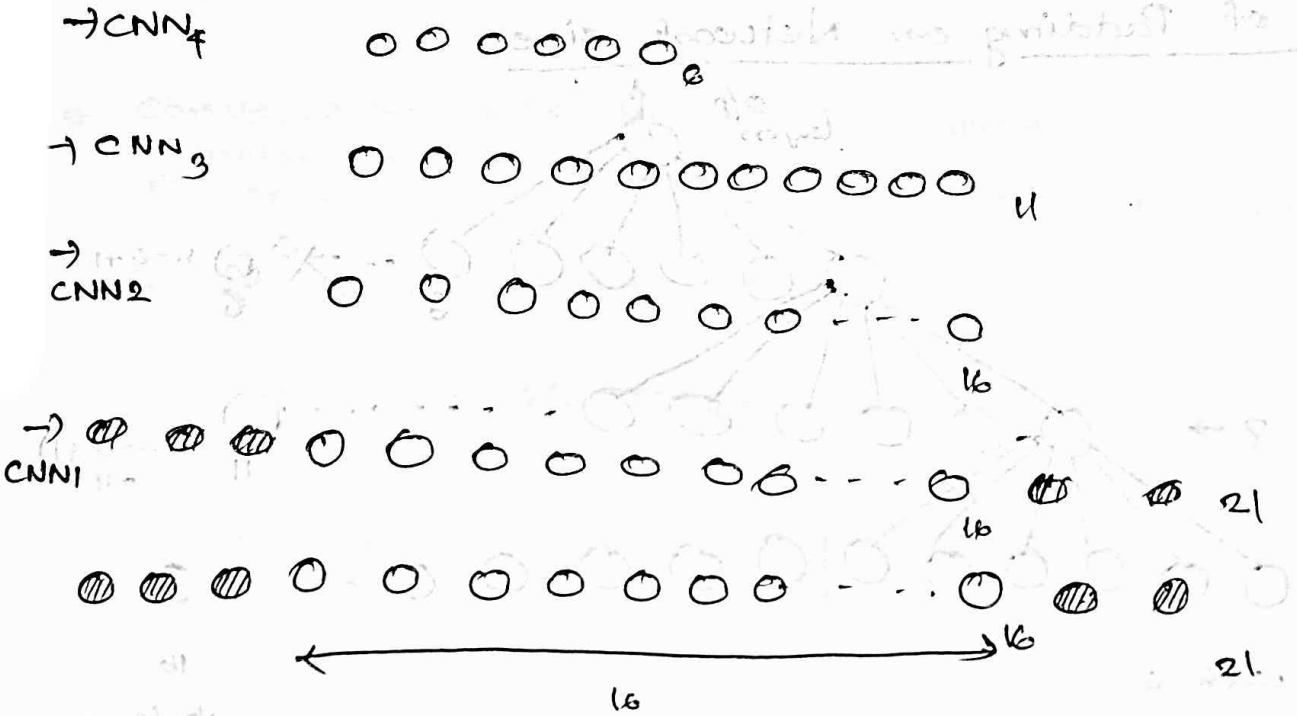
→ Valid (no padding)
→ Half
→ Full
→ Same

Effect of Padding on Network size



In this example we do not use pooling the convolutions operations itself shrink the nlw size. by 5 pixels at each layer so only 2 CNN layers is possible in this case solution to above problem is to use padded convolution.

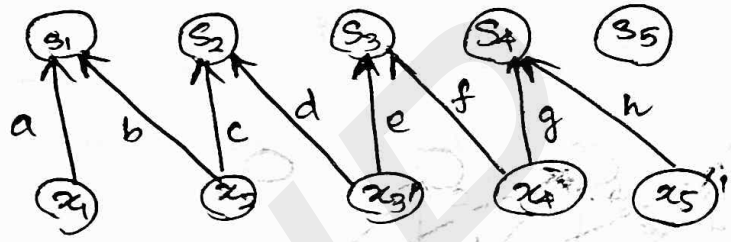


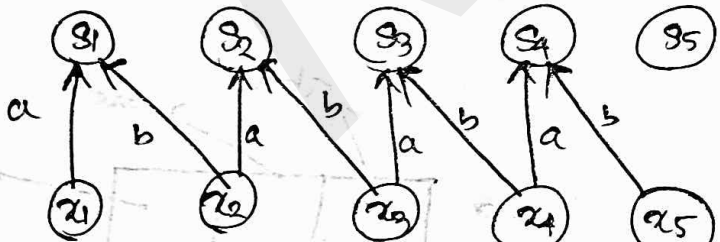


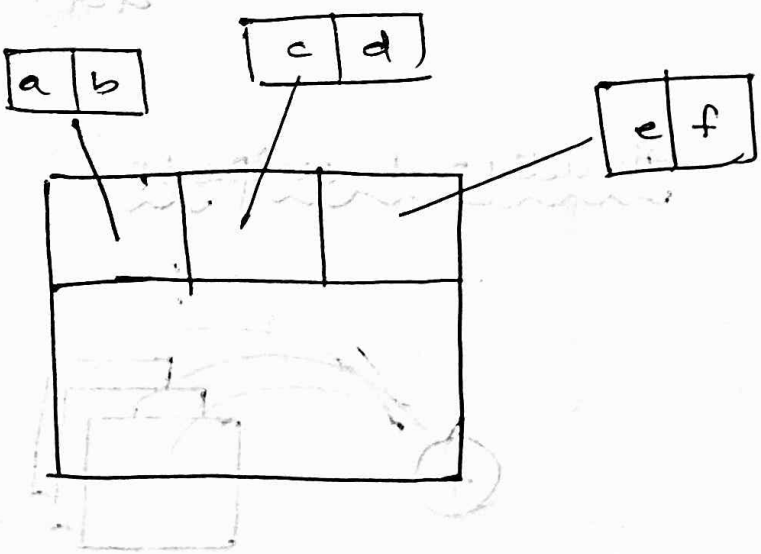
After performing padding on each layer the number of CNN layers that can be added to the network is increased.

~~unshared~~

Unshared Convolution



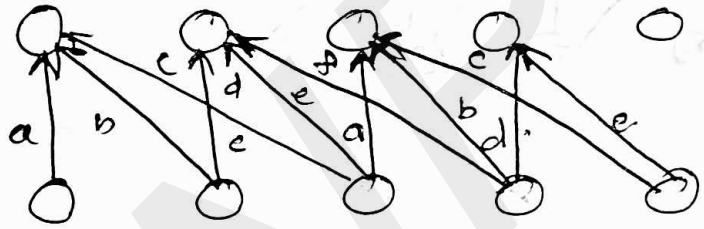


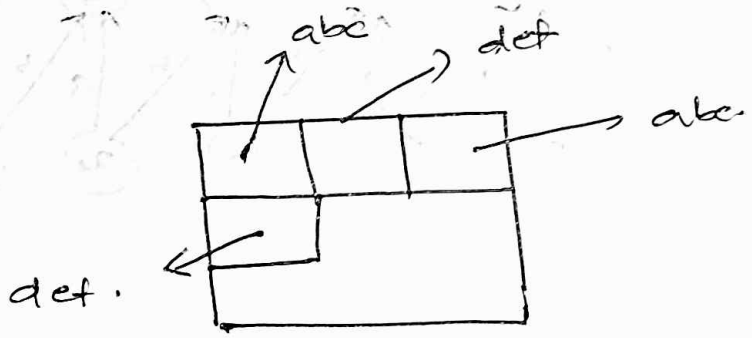


In an unshared convolution different parameters are used at each convolution.

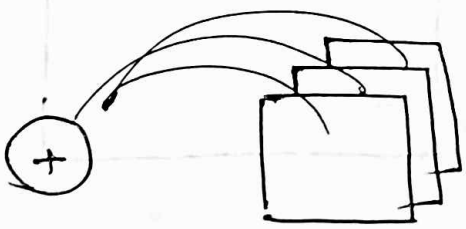
~~Structured Outputs~~

Tiled convolution





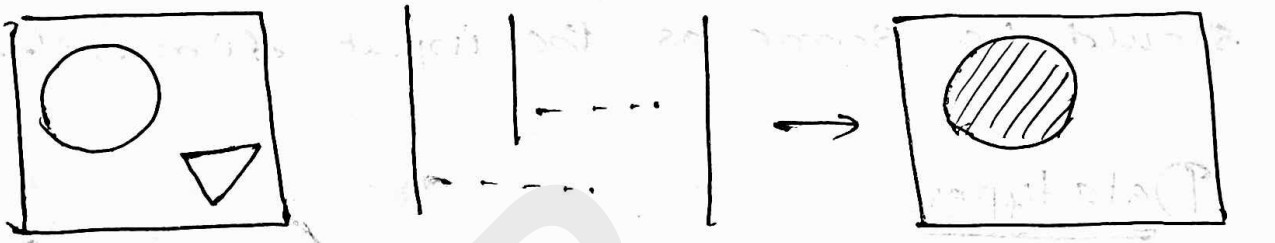
Structured outputs



convolution neural network can be used to output a high dimensional structured objects rather than just predicting the class label for a classification task or a real value for a regression task.

Segmentation Circle

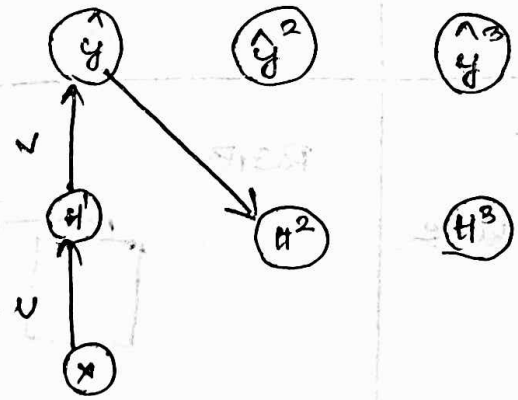
eg:-



300×200 300×200

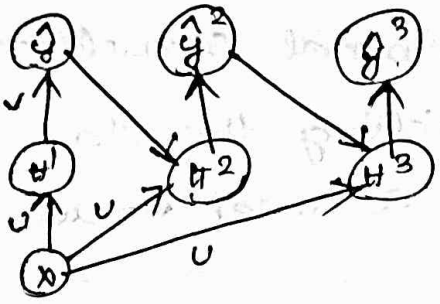
In such task the model needs to label every pixel in an image and draw precise the mask that follow the outline of individual objects

Example:— Architecture.



x → I/P pixel

y → corresponding label



Process repeat until reach the correct label.

One issue need to handle is output dimension should be same as the input dimension.

Datatypes

~~a) One dimensional data.~~

Single channel	Multi channel
ONE DIMENSIONAL DATA. a) Audio data Amplitude Time.	a) Animation data
TWO DIMENSIONAL DATA. Grey scale / Black and white H W	RGB
THREE DIMENSIONAL DATA. H W d Volumetric data eg:- CT scans image etc	Colour Vedio data